Supplementary Notes: The Hyperplane Theorems and the Minimax Theorem

STAT 220B

1 Introduction

These notes accompany Lecture 7 and provide full proofs of the material on supporting and separating hyperplanes (Berger §5.2.5) and the Minimax Theorem (Berger §5.2.6). Berger's text develops these results across roughly seven pages, with several lemmas left as exercises. The exposition here is self-contained: every claim is proved, every assumption is identified, and the geometric intuition is made explicit alongside the algebra.

The reader is assumed to be familiar with the S-game reformulation from Lecture 7. Briefly: given a two-person zero-sum game with finite parameter space $\Theta = \{\theta_1, \dots, \theta_m\}$, the risk set is the subset of $\mathbb{R}^m$

$$S = \{R(\delta^*) : \delta^* \in \mathcal{A}^*\}, \qquad R(\delta^*) = \big(L(\theta_1, \delta^*), \dots, L(\theta_m, \delta^*)\big)^T,$$

where $\mathcal{A}^*$ is the set of randomized actions available to Player II. The risk set is convex (Lemma 3 of Lecture 7). The game value, when it exists, equals the smallest α such that the closed lower quadrant $Q_\alpha = \{z \in \mathbb{R}^m : z_i \le \alpha \text{ for all } i\}$ meets S.

Throughout these notes, all vectors are column vectors, $\xi^T z$ is the standard inner product on $\mathbb{R}^m$, and $\|z\| = (z^T z)^{1/2}$. We say S is **bounded below** if there exists a finite K with $s_i \ge -K$ for all $s \in S$ and all coordinates i; this is automatic from the standing assumption that the loss L is bounded below.

2 Hyperplanes and convex sets

2.1 Definitions and elementary facts

Definition 2.1. A **hyperplane** in $\mathbb{R}^m$ is a set of the form

$$H(\xi, k) = \{z \in \mathbb{R}^m : \xi^T z = k\},$$

where $\xi \in \mathbb{R}^m$ is a nonzero vector and $k \in \mathbb{R}$. The vector ξ is the **normal** to the hyperplane.

Geometrically, $H(\xi, k)$ is the affine subspace of dimension $m - 1$ perpendicular to ξ and translated from the origin to contain any point z_0 with $\xi^T z_0 = k$. The two open half-spaces

$$H^+(\xi, k) = \{z : \xi^T z > k\}, \qquad H^-(\xi, k) = \{z : \xi^T z < k\},$$

together with $H(\xi, k)$ itself, partition $\mathbb{R}^m$ into three disjoint pieces.

Definition 2.2. A hyperplane $H(\xi, k)$ **supports** a set $S \subseteq \mathbb{R}^m$ at a point $s_0 \in S$ if $s_0 \in H(\xi, k)$ and $\xi^T s \geq k$ for all $s \in S$.

Definition 2.3. Two sets $S_1, S_2 \subseteq \mathbb{R}^m$ are **separated** by a hyperplane $H(\xi, k)$ if $\xi^T s_1 \geq k$ for all $s_1 \in S_1$ and $\xi^T s_2 \leq k$ for all $s_2 \in S_2$.

The reader should keep two pictures in mind. First, a convex set has a supporting hyperplane at each of its boundary points: imagine pressing a flat ruler against a convex blob from outside. Second, two disjoint convex sets can always be peeled apart by a flat hyperplane: imagine sliding a sheet of paper between two non-overlapping convex blobs.

2.2 Closest-point lemma

The proofs of both hyperplane theorems rest on the following lemma about projections onto closed convex sets.

Lemma 2.4. *Let $C \subseteq \mathbb{R}^m$ be a nonempty closed convex set, and let $y \in \mathbb{R}^m$ with $y \notin C$. Then there exists a unique point $s^* \in C$ minimizing $\|s - y\|$ over $s \in C$. Moreover, for every $s \in C$,*

$$(s^* - y)^T (s - s^*) \geq 0. \tag{†}$$

Proof. Existence. Pick any $s_0 \in C$, and let $r = \|s_0 - y\|$. The set $C \cap \overline{B}(y, r)$, where $\overline{B}(y, r) = \{z : \|z - y\| \leq r\}$ is the closed ball of radius r around y, is nonempty, closed, and bounded. Any minimizer of $\|s - y\|$ over C must lie in this set. By compactness and continuity, a minimizer $s^* \in C \cap \overline{B}(y, r)$ exists.

Uniqueness. Suppose $s_1, s_2 \in C$ both minimize, with $\|s_i - y\| = d$. The midpoint $\bar{s} = (s_1 + s_2)/2$ lies in C by convexity. By the parallelogram identity,

$$\|s_1 - y\|^2 + \|s_2 - y\|^2 = 2\|\bar{s} - y\|^2 + \tfrac{1}{2}\|s_1 - s_2\|^2.$$

Substituting $\|s_i - y\|^2 = d^2$: $\|\bar{s} - y\|^2 = d^2 - \frac{1}{4}\|s_1 - s_2\|^2$. If $s_1 \neq s_2$, the right side is strictly less than d^2, contradicting minimality. Hence $s_1 = s_2$.

Variational inequality. Fix $s \in C$ and $t \in [0, 1]$. The point $s_t = (1 - t)s^* + ts$ lies in C by convexity, so

$$\|s^* - y\|^2 \leq \|s_t - y\|^2 = \|s^* - y\|^2 + 2t(s^* - y)^T (s - s^*) + t^2 \|s - s^*\|^2.$$

Dividing by $t > 0$ and letting $t \downarrow 0$ gives $0 \leq 2(s^* - y)^T (s - s^*)$, which is (†). □

The variational inequality (†) is the heart of the matter: at the closest point s^*, the vector from y to s^* makes a non-acute angle with the vector from s^* to any other point of C. The hyperplane through s^* normal to $s^* - y$ is the supporting hyperplane we seek.

3 The Supporting Hyperplane Theorem

Theorem 3.1 (Supporting Hyperplane Theorem)**.** *Let $S \subseteq \mathbb{R}^m$ be a nonempty convex set, and let s_0 be a boundary point of S. Then there exists a nonzero $\xi \in \mathbb{R}^m$ such that*

$$\xi^T s \geq \xi^T s_0 \quad \text{for all } s \in S.$$

Equivalently, the hyperplane $H(\xi, \xi^T s_0)$ supports S at s_0.

Proof. Since s_0 is a boundary point of S, there exists a sequence $y_n \in \mathbb{R}^m \setminus \overline{S}$ with $y_n \to s_0$. The set $\overline{S}$ is closed and convex. By Lemma 2.4 applied to $C = \overline{S}$ and $y = y_n$, there is a unique $s_n^* \in \overline{S}$ minimizing $\|s - y_n\|$ over $s \in \overline{S}$, and

$$(s_n^* - y_n)^T(s - s_n^*) \geq 0 \quad \text{for all } s \in \overline{S}. \tag{‡}$$

Set $\xi_n = (s_n^* - y_n)/\|s_n^* - y_n\|$. The denominator is positive since $y_n \notin \overline{S}$, and $\|\xi_n\| = 1$. Inequality (‡) divides through to give

$$\xi_n^T(s - s_n^*) \geq 0 \quad \text{for all } s \in \overline{S}. \tag{‡$'$}$$

By Bolzano–Weierstrass, the bounded sequence $\{\xi_n\}$ has a convergent subsequence $\xi_{n(k)} \to \xi$ with $\|\xi\| = 1$, so $\xi \neq 0$. Since $\|s_n^* - y_n\| \leq \|s_0 - y_n\| \to 0$, we also have $s_n^* \to s_0$.

Take any $s \in S$. From (‡$'$) along the subsequence, $\xi_{n(k)}^T(s - s_{n(k)}^*) \geq 0$. Passing to the limit, $\xi^T(s - s_0) \geq 0$, i.e., $\xi^T s \geq \xi^T s_0$. □

Remark 3.2. The supporting hyperplane is not unique in general: at a corner of a polytope, there is a whole family of supporting hyperplanes. Uniqueness holds at smooth boundary points but not at kinks. This non-uniqueness reflects the existence of multiple optimal strategies in some games.

4 The Separating Hyperplane Theorem

Theorem 4.1 (Separating Hyperplane Theorem)**.** *Let $S_1, S_2 \subseteq \mathbb{R}^m$ be disjoint nonempty convex sets. Then there exists a nonzero $\xi \in \mathbb{R}^m$ and a scalar k such that*

$$\xi^T s_1 \geq k \textit{ for all } s_1 \in S_1, \qquad \xi^T s_2 \leq k \textit{ for all } s_2 \in S_2. \tag{$\star$}$$

Proof. Case 1: $\overline{S_1} \cap \overline{S_2} = \emptyset$ and one of them is bounded. Without loss of generality, suppose $\overline{S_2}$ is bounded (compact). The difference set $D = \overline{S_1} - \overline{S_2}$ is convex (Minkowski sum of convex sets) and closed (closed plus compact is closed). Disjointness gives $0 \notin D$.

Apply Lemma 2.4 with $C = D$ and $y = 0$: there exists $d^* \in D$ with $d^* \neq 0$ and $(d^*)^T(d - d^*) \geq 0$ for all $d \in D$. Setting $\xi = d^*$, this gives $\xi^T d \geq \|\xi\|^2 > 0$ for all $d \in D$. Writing $d = s_1 - s_2$,

$$\xi^T s_1 - \xi^T s_2 \geq \|\xi\|^2 \quad \text{for all } s_1 \in \overline{S_1}, s_2 \in \overline{S_2}.$$

Any k between $\sup_{s_2} \xi^T s_2$ and $\inf_{s_1} \xi^T s_1$ gives ($\star$).

Case 2: General case. Consider $C = S_1 - S_2$. Since S_1 and S_2 are disjoint and convex, C is convex and $0 \notin C$. Either $0 \notin \overline{C}$ (reduce to Case 1) or 0 is a boundary point of $\overline{C}$. In the latter case, Theorem 3.1 applied to $\overline{C}$ at 0 yields nonzero ξ with $\xi^T c \geq 0$ for all $c \in \overline{C}$, hence $\xi^T s_1 \geq \xi^T s_2$ for all $s_1 \in S_1, s_2 \in S_2$. Setting $k = \sup_{s_2 \in S_2} \xi^T s_2$ completes the proof. □

Remark 4.2. The separation is **weak**: the inequalities are not strict, so S_1 and S_2 may share boundary points on the separating hyperplane. **Strict separation** requires additional assumptions (one set closed, the other compact, with disjointness). The weak version suffices for our applications.

5 Geometry of the risk set

We now apply the hyperplane theorems to the risk set S. Recall:

$$Q_\alpha = \{z \in \mathbb{R}^m : z_i \le \alpha \ \forall i\}, \qquad S_\alpha = S \cap Q_\alpha, \qquad \alpha_M = \inf\{\alpha : S_\alpha \ne \emptyset\}.$$

Boundedness below gives $\alpha_M > -\infty$; nonemptiness of S gives $\alpha_M < \infty$. The quantity α_M is the candidate value of the game.

Definition 5.1. A point $s \in \overline{S}$ is a **lower boundary point** of S if there is no other $s' \in \overline{S}$ with $s' \le s$ coordinate-wise (that is, $s'_i \le s_i$ for all i, with at least one strict inequality). The set of lower boundary points is denoted $\lambda(S)$.

The set $\lambda(S)$ is the southwest boundary of $\overline{S}$ in coordinate-wise partial order. A risk point not in $\lambda(S)$ corresponds to an inadmissible strategy.

Lemma 5.2. *Suppose S is bounded below and closed from below (that is, $\lambda(S) \subseteq S$). Let $s^0 \in \lambda(S)$. Then there exists $\pi^0 \in \Omega_m$ (probability vector) with*

$$(\pi^0)^T s \ge (\pi^0)^T s^0 \quad \textit{for all } s \in S. \tag{$\sharp$}$$

Proof. Let $T = \{z \in \mathbb{R}^m : z_i < s_i^0 \ \forall i\}$. The set T is convex and open. If some $s \in S$ had $s_i < s_i^0$ for all i, then $s \le s^0$ with strict inequalities, contradicting $s^0 \in \lambda(S)$. So $T \cap S = \emptyset$.

Apply Theorem 4.1 to S and T: there is nonzero ξ and k with $\xi^T s \ge k$ for $s \in S$ and $\xi^T z \le k$ for $z \in T$.

We claim $\xi_i \ge 0$. If $\xi_j < 0$, sending $z_j \to -\infty$ drives $\xi^T z \to +\infty$, contradicting $\xi^T z \le k$. The supremum of $\xi^T z$ over $z \in T$ is $\xi^T s^0$, so $k \ge \xi^T s^0$. Evaluating at $s = s^0 \in S$ gives $k \le \xi^T s^0$. Hence $k = \xi^T s^0$.

Since $\xi \ge 0$ and $\xi \ne 0$, $\sum_i \xi_i > 0$. Set $\pi^0 = \xi / \sum_i \xi_i \in \Omega_m$, and ($\sharp$) holds. ☐

Lemma 5.3. *Suppose S is bounded below and closed from below. For every $s \in S$, either $s \in \lambda(S)$, or there exists $s' \in \lambda(S)$ with $s' \le s$ coordinate-wise and $s' \ne s$.*

Proof. If $s \in \lambda(S)$ we are done. Otherwise, there is $s^{(1)} \in \overline{S}$ with $s^{(1)} \le s$, strict in some coordinate. Iterate: at stage n, either $s^{(n)} \in \lambda(S)$ and we stop, or there is $s^{(n+1)} \in \overline{S}$ with $s^{(n+1)} \le s^{(n)}$, strict in some coordinate. Each coordinate sequence is monotone decreasing, bounded below by $-K$, hence converges. Let $s^* = \lim_n s^{(n)} \in \overline{S}$. One shows $s^* \in \lambda(S) \subseteq S$, so we take $s' = s^*$. ☐

The argument is a "descend until you can't" argument in a closed lower-bounded set. The output: $\lambda(S)$ is rich enough to dominate every other point of S coordinate-wise.

6 The Minimax Theorem

We now state and prove the main result.

Theorem 6.1 (Minimax Theorem, Berger Theorem 15)**.** *Consider a two-person zero-sum game with finite $\Theta = \{\theta_1, \dots, \theta_m\}$ and bounded loss $L \ge -K$. Let S be the risk set, and $\alpha_M = \inf\{\alpha : S_\alpha \ne \emptyset\}$. Then:*

(i) The game has a value V, and $V = \alpha_M$. That is,

$$\inf_{\delta^* \in \mathcal{A}^*} \sup_{\pi \in \Omega_m} L(\pi, \delta^*) = \sup_{\pi \in \Omega_m} \inf_{\delta^* \in \mathcal{A}^*} L(\pi, \delta^*) = \alpha_M.$$

(ii) A maximin strategy $\pi^M \in \Omega_m$ for Player I exists.

(iii) If S is closed from below, a minimax strategy $\delta^{*M} \in \mathcal{A}^*$ for Player II exists, and $L(\pi^M, \delta^{*M}) = V$.

The proof proceeds in seven steps. We assume S is bounded below throughout and add "closed from below" only at the final step.

6.1 Step 1: Separating the risk set from the lower quadrant

Define the open set

$$T = \{z \in \mathbb{R}^m : z_i < \alpha_M \ \forall i\}.$$

This is the interior of Q_{α_M}, and is nonempty.

We claim $S \cap T = \emptyset$. If some $s \in S$ had $s_i < \alpha_M$ for all i, let $\alpha = \max_i s_i < \alpha_M$. Then $s \in S_\alpha$, contradicting the definition of α_M.

Both S and T are convex. By the Separating Hyperplane Theorem (Theorem 4.1), there exists nonzero ξ and scalar k with

$$\xi^T s \geq k \ \forall s \in S, \qquad \xi^T z \leq k \ \forall z \in T. \qquad (\star_1)$$

6.2 Step 2: The normal vector has nonnegative entries

We claim $\xi_i \geq 0$ for every i. Suppose $\xi_j < 0$ for some j. Fix $z^0 \in T$. For each $t > 0$, the point $z^t = z^0 - te_j$ still lies in T, but

$$\xi^T z^t = \xi^T z^0 - t\xi_j \to +\infty \quad \text{as } t \to \infty,$$

contradicting $\xi^T z \leq k$ uniformly.

6.3 Step 3: Rescaling the normal to a probability vector

The supremum of $\xi^T z$ over $z \in T$ is achieved in the limit by sending each $z_i \uparrow \alpha_M$:

$$\sup_{z \in T} \xi^T z = \alpha_M \sum_{i=1}^{m} \xi_i.$$

Combined with the second inequality in $(\star_1)$, $k \geq \alpha_M \sum_i \xi_i$. For any $s \in S$ with $\max_i s_i$ close to α_M, $\xi^T s \leq (\sum_i \xi_i)\alpha_M(1 + \epsilon)$ for small ϵ, while $\xi^T s \geq k$. Hence $k = \alpha_M \sum_i \xi_i$.

Since $\xi \geq 0$ and $\xi \neq 0$, $\sum_i \xi_i > 0$. Define $\pi = \xi / \sum_i \xi_i \in \Omega_m$. Dividing $\xi^T s \geq k$ by $\sum_i \xi_i$,

$$\pi^T s \geq \alpha_M \quad \text{for all } s \in S. \qquad (\star_2)$$

6.4 Step 4: A lower bound on the lower value

The lower value is $\underline{V} = \sup_{\pi' \in \Omega_m} \inf_{s \in S} (\pi')^T s$. The probability vector π satisfies $\inf_{s \in S} \pi^T s \geq \alpha_M$ by $(\star_2)$, so

$$\underline{V} \geq \inf_{s \in S} \pi^T s \geq \alpha_M. \qquad (\star_3)$$

6.5 Step 5: An upper bound on the upper value

The upper value is

$$\overline{V} = \inf_{s \in S} \sup_{\pi' \in \Omega_m} (\pi')^T s = \inf_{s \in S} \max_i s_i,$$

since the supremum of a linear functional over Ω_m is the maximum coordinate.

By definition of α_M, for every $\epsilon > 0$ there is $s^\epsilon \in S$ with $\max_i s_i^\epsilon \leq \alpha_M + \epsilon$. So $\overline{V} \leq \alpha_M + \epsilon$. Letting $\epsilon \downarrow 0$,

$$\overline{V} \leq \alpha_M. \qquad (\star_4)$$

6.6 Step 6: Value exists and π is maximin

Combining $\underline{V} \leq \overline{V}$ (Lemma 2 of Lecture 6) with $(\star_3)$ and $(\star_4)$,

$$\alpha_M \leq \underline{V} \leq \overline{V} \leq \alpha_M,$$

so $\underline{V} = \overline{V} = \alpha_M = V$. The game has a value, and $V = \alpha_M$. This proves (i).

The probability vector π from Step 3 achieves $\inf_{s \in S} \pi^T s \geq \alpha_M = \underline{V}$, so it is a maximin strategy. Set $\pi^M = \pi$. This proves (ii).

6.7 Step 7: Existence of a minimax strategy when S is closed from below

From Step 5, for each integer $n \geq 1$ there is $s^{(n)} \in S$ with $\max_i s_i^{(n)} \leq \alpha_M + 1/n$. The sequence is contained in the closed bounded box $\{z : -K \leq z_i \leq \alpha_M + 1\}$. By Bolzano-Weierstrass, $s^{(n_k)} \to s^*$ for some subsequence, with $s^* \in \overline{S}$. Continuity gives $\max_i s_i^* \leq \alpha_M$, while $(\star_2)$ applied to s^* via π forces $\max_i s_i^* \geq \pi^T s^* \geq \alpha_M$. So $\max_i s_i^* = \alpha_M$.

Case A: $s^* \in \lambda(S)$. Since S is closed from below, $s^* \in S$, so $s^* \in S_{\alpha_M}$ is a minimax risk point.

Case B: $s^* \notin \lambda(S)$. By Lemma 5.3, there exists $s' \in \lambda(S) \subseteq S$ with $s' \leq s^*$ coordinate-wise, $s' \neq s^*$. Then $\max_i s_i' \leq \max_i s_i^* = \alpha_M$, and as above $\max_i s_i' = \alpha_M$, so $s' \in S_{\alpha_M}$ is a minimax risk point.

In either case, a minimax risk point exists in S, and the corresponding randomized strategy is a minimax strategy δ^{*M}. The product $L(\pi^M, \delta^{*M}) = \sum_i \pi_i^M L(\theta_i, \delta^{*M})$ satisfies $L(\theta_i, \delta^{*M}) \leq V$ for each i, while $(\star_2)$ forces the weighted average to be at least $\alpha_M = V$. Therefore $L(\pi^M, \delta^{*M}) = V$, completing the proof of (iii). $\square$

7 Remarks and consequences

7.1 What the proof establishes

The proof is constructive. Given the risk set S:

(a) Identify α_M by shrinking Q_α until it just touches S.

(b) Find a supporting hyperplane of S at the point of tangency; its normal direction (after rescaling to a probability vector) is the maximin strategy π^M.

(c) The minimax strategy δ^{*M} is any randomized strategy with risk point in $S_{\alpha_M} = S \cap Q_{\alpha_M}$.

For finite games, S is a polytope and steps (a) to (c) reduce to linear programming. This is no accident: the Minimax Theorem is the duality theorem of linear programming in disguise.

7.2 Tangency and the maximin strategy

At the minimax risk point $s^M \in S_{\alpha_M}$, the supporting hyperplane of S has normal direction π^M:

$$H(\pi^M, V) = \{z \in \mathbb{R}^m : (\pi^M)^T z = V\}$$

passes through s^M and has all of S on or above it. This tangency is the geometric content of Theorem 1 of Lecture 6, which asserts that a minimax strategy is Bayes against the maximin prior. The maximin prior is precisely the supporting hyperplane's normal at s^M, and the Bayes rule against π^M minimizes $\sum_i \pi_i^M s_i$ over $s \in S$, which is the linear functional whose level set is the supporting hyperplane.

7.3 What if Θ is infinite?

The proof uses finiteness of Θ in two essential places. First, $S \subseteq \mathbb{R}^m$ has finite ambient dimension, so the hyperplane theorems apply directly. Second, Bolzano-Weierstrass extracts a convergent subsequence.

For infinite Θ, S becomes a subset of an infinite-dimensional function space, and the Bolzano-Weierstrass argument fails. Existence of minimax strategies then requires topological assumptions on Θ and $\mathcal{A}$; this is the content of Berger Theorem 16 and more general results by Wald, Le Cam, and Blackwell-Girshick. We do not pursue these here.

7.4 Connection to linear programming duality

A two-person zero-sum game with finite Θ and finite $\mathcal{A}$ corresponds to a pair of dual linear programs. Player II solves

$$\min_{(\delta^*, v)} v \;\text{ s.t. }\; \sum_a L(\theta_i, a)\delta^*(a) \le v \;\forall i,\; \sum_a \delta^*(a) = 1,\; \delta^* \ge 0,$$

and Player I solves the dual

$$\max_{(\pi, w)} w \;\text{ s.t. }\; \sum_i L(\theta_i, a)\pi(\theta_i) \ge w \;\forall a,\; \sum_i \pi(\theta_i) = 1,\; \pi \ge 0.$$

The Minimax Theorem says the two optimal values coincide, which is precisely strong LP duality. The constructive proof above is essentially the simplex algorithm in geometric language.

8 Worked example: Berger Example 11

To make the proof concrete, we revisit Berger's Example 11 from Lecture 7. The loss matrix is

$$L = \begin{pmatrix} 2 & 2 & 0 & 0.5 \\ 0 & 1 & 2 & 1 \end{pmatrix},$$

with $\Theta = \{\theta_1, \theta_2\}$ and $\mathcal{A} = \{a_1, a_2, a_3, a_4\}$. The columns of L are the four nonrandomized risk points in $\mathbb{R}^2$:

$$R(a_1) = \begin{pmatrix} 2 \\ 0 \end{pmatrix},\; R(a_2) = \begin{pmatrix} 2 \\ 1 \end{pmatrix},\; R(a_3) = \begin{pmatrix} 0 \\ 2 \end{pmatrix},\; R(a_4) = \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}.$$

The risk set S is the convex hull.

Locating α_M. The lower boundary $\lambda(S)$ runs from $R(a_3) = (0, 2)$ to $R(a_4) = (0.5, 1)$ to $R(a_1) = (2, 0)$. On the segment from $(0.5, 1)$ to $(2, 0)$, parametrize as $(2 - 1.5t, t)$ for $t \in [0, 1]$. We want $\max\{2 - 1.5t, t\}$ minimized, which occurs when $2 - 1.5t = t$, i.e., $t = 0.8$. The minimax risk point is $s^M = (0.8, 0.8)$ with $\alpha_M = 0.8$.

Finding the supporting hyperplane. The segment direction is $(1.5, -1)$. The inward normal (into the interior, where S lies) is in direction $(1, 1.5)$. Normalizing, $\pi = (1, 1.5)/2.5 = (0.4, 0.6)$, so $\pi^M = (0.4, 0.6)^T$.

Verifying $(\star_2)$. For any $s = (2 - 1.5t, t)$ on the segment,

$$\pi^T s = 0.4(2 - 1.5t) + 0.6t = 0.8 - 0.6t + 0.6t = 0.8 = \alpha_M.$$

The supporting hyperplane contains the entire segment. Check on the other vertices: $\pi^T R(a_2) = 0.4(2) + 0.6(1) = 1.4 \geq 0.8$, $\pi^T R(a_3) = 0.4(0) + 0.6(2) = 1.2 \geq 0.8$. So $\pi^T s \geq 0.8$ over S.

Finding the minimax strategy. The point $s^M = (0.8, 0.8)$ lies on the segment from $R(a_4) = (0.5, 1)$ to $R(a_1) = (2, 0)$. Solving $0.8 = 0.5\tau + 2(1 - \tau) = 2 - 1.5\tau$ gives $\tau = 0.8$. The minimax strategy plays a_4 with probability 0.8 and a_1 with probability 0.2.

Verification using Theorem 2 of Lecture 6.

$$L(\theta_1, \delta^{*M}) = 0.8(0.5) + 0.2(2) = 0.8, \qquad L(\theta_2, \delta^{*M}) = 0.8(1) + 0.2(0) = 0.8.$$

$$L(\pi^M, a_1) = 0.4(2) + 0.6(0) = 0.8, \quad L(\pi^M, a_4) = 0.4(0.5) + 0.6(1) = 0.8.$$

And $L(\pi^M, a_2) = 1.4 \geq 0.8$, $L(\pi^M, a_3) = 1.2 \geq 0.8$. The condition $L(\theta, \delta^{*M}) \leq 0.8 \leq L(\pi^M, a)$ holds for all θ, a, confirming that π^M, δ^{*M} are optimal with value $V = 0.8$.

9 Summary

The Minimax Theorem is the existence guarantee underwriting the minimax principle. Its proof has three conceptual ingredients:

1. The S-game reformulation (Lecture 7) reduces the game to a geometric problem in $\mathbb{R}^m$.
2. The hyperplane theorems provide the analytical mechanism for relating the lower quadrant Q_{α_M} to the risk set S.
3. Bolzano-Weierstrass extracts a minimax risk point from a sequence of approximate optima.

The proof is constructive: given S, the maximin strategy is the normal to the supporting hyperplane at the minimax risk point, rescaled to a probability vector. For finite games this reduces to linear programming.

These notes treat only finite Θ. For continuous parameter spaces, additional structural assumptions are needed (Berger Theorem 16). The geometric intuition persists: minimax strategies are supported on the worst-case sets for the adversary, and the tangent-hyperplane picture applies via approximation.

Further reading. Berger, *Statistical Decision Theory and Bayesian Analysis*, §5.2.4 to §5.2.6. Rockafellar, *Convex Analysis* (Princeton, 1970), Chapters 11 to 14 for hyperplane theorems in convex analysis. Boyd and Vandenberghe, *Convex Optimization* (Cambridge, 2004), §5.2 for the LP duality connection.